

Final Presentation

Stock Direction Prediction



Team 003: Itay Mutzafi | Moran Zaks | Shaked Schnarch

Reminder

- Foundations & Baseline (Pre-Checkpoint)
- Roadmap And Progress
- Benchmarks and Success Criteria

Recent Progress And Main Challenges

- Feature Engineering
- “Model Zoo” Experiment- Framework
- “Model Zoo” Experiment- Results
- Feature Selection
- Bull & Bear regimes

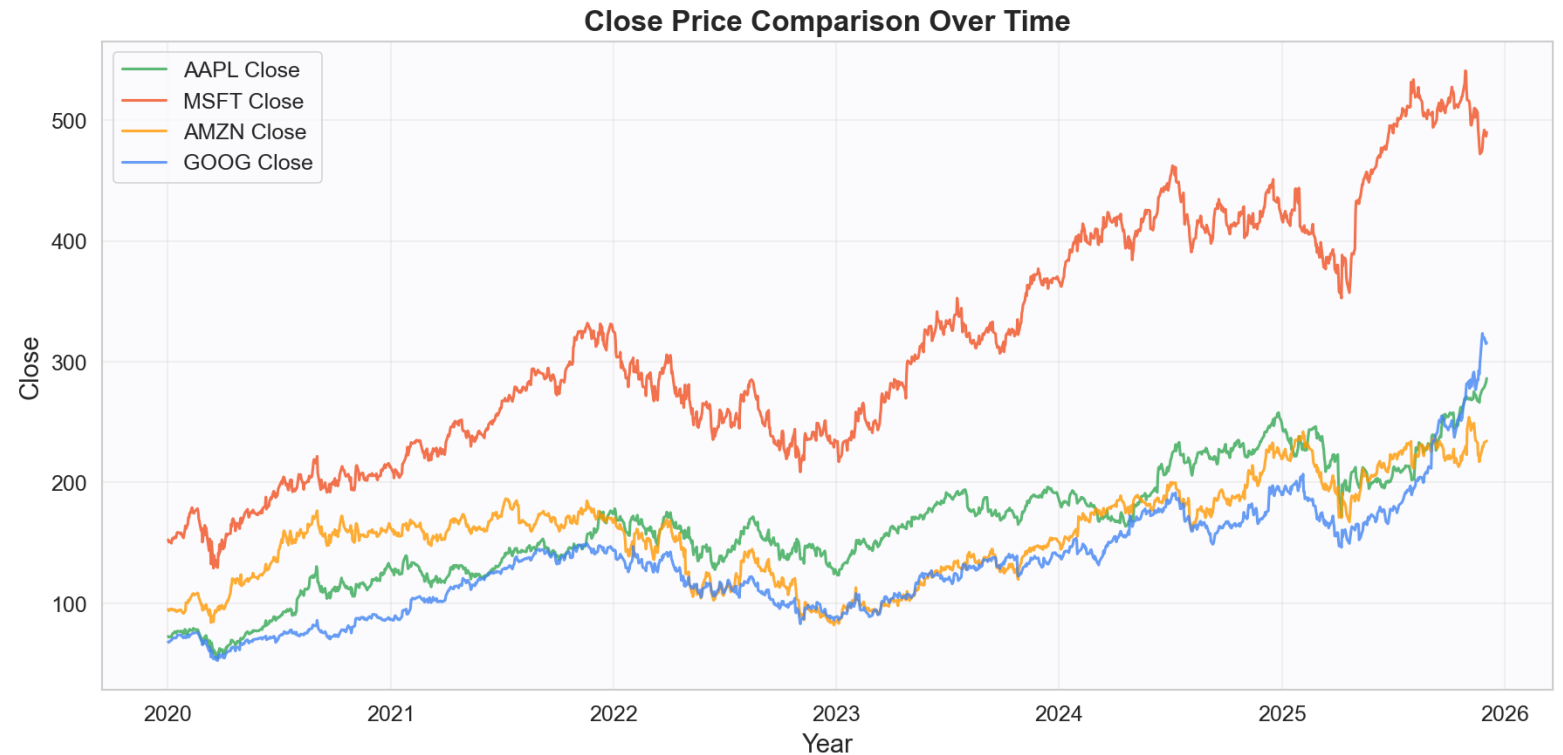
Project status

- Project status – Finished tasks
- Project status – Future Directions

Reminder



- **The Task:** Directional of next-day Logarithmic Returns
- **Target Universe:** Mega-cap technology stocks: AAPL, MSFT, AMZN, and GOOG
- **Data:** Yahoo Finance, News



Foundations & Baseline (Pre-Checkpoint)

- **Problem Formulation:** Defined Multi-Task objectives (Regression & Classification) and KPIs
- **Core Infrastructure:** Initialized repository and modular code architecture (src)
- **EDA and Profiling:** Exploratory Data Analysis and statistical profiling of returns
- **Resource Refinement:** Optimized news scraping and FinBERT preprocessing pipelines
- **Baseline Engineering:** Constructed primary OHLCV lags and technical indicators

Final Project Report: Stock Market Prediction

Authors: Team 003 Itay Mutzafi, Moran Zaks, Shaked Schriach
Course: Workshop in Data Science (0368-3528), Tel Aviv University
Date: October 2025 - March 2026

Abstract

This report presents the methodology and empirical findings of a final capstone project aimed at predicting short-term directional movements of major technology indices (AAPL, MSFT, AMZN, GOOG) stationarity in financial time series, this study focuses on forecasting **Directional Movements** based on Logarithmic Returns.

The project employs a modular code architecture (SFC) and integrates:

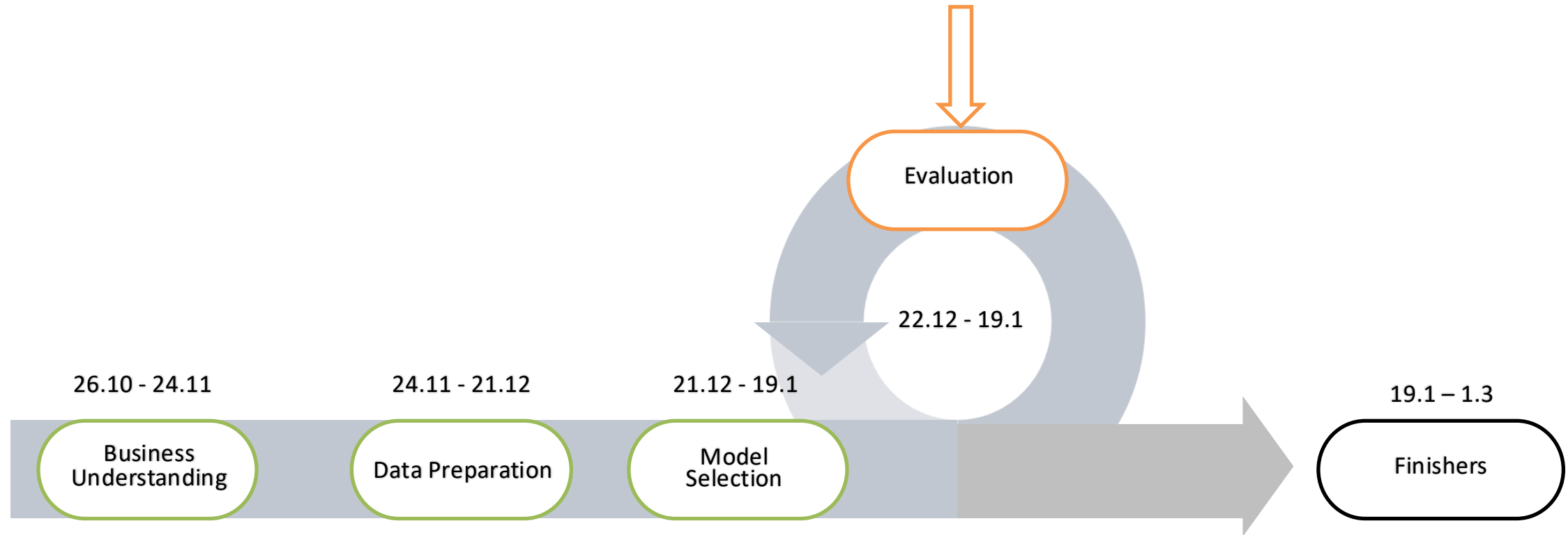
1. **Feature Engineering:** A robust synthesis of technical indicators (RSI, MACD, Bollinger Bands) and lagged variables to capture market momentum and trend dynamics.
2. **Walk-Forward Validation:** Implementation of a strict temporal validation mechanism that mimics real-time trading environments, thereby eliminating look-ahead bias and data leakage.
3. **Model Selection:** An empirical comparison between a Baseline (Dummy) model, Logistic Regression, and Random Forest classifiers.

Results indicate that the selected model achieved a **Sharpe Ratio** superior to the baseline, demonstrating the viability of technical signal-based strategies for enhancing risk-adjusted returns.

Table of Contents

- 1. Introduction
 - 1.1 Business Understanding
 - 1.1.1 Research Objective
 - 1.1.2 The Scientific Challenge
 - 1.1.3 Theoretical Background
 - 1.2 Problem Formulation
 - 1.2.1 Objective and Goals
 - 1.2.2 Key Performance Indicators (KPIs)
 - 1.2.3 Target Feature
 - 1.2.3.1 Statistical Properties and Stationarity Tests
 - 1.2.3.2 Feasibility Analysis and Signal Exploration
 - 1.2.3.3 Target Variable Final Definition (Log>Returns)
 - 1.2.4 Methodology and Validation Strategy
 - 1.2.4.1 Evaluation Metrics Definition
 - 1.2.4.2 Validation Schema: Strict Walk-Forward Split
- 2. Data Access
 - 2.1 Data Ingestion Verification
 - 2.2 Day and Month Features
- 3. Exploratory Data Analysis (EDA)
 - 3.1 Correlation
 - 3.2 Basic Features
 - 3.3 Seasonality

Roadmap And Progress



Benchmarks and Success Criteria



The "Accuracy Ceiling" in Financial ML:

- **Industry Standard (Kaggle/Jane Street):** Top-tier models achieve **52% – 55%** accuracy
- **Zhong & Enke:** ~60% considered near the upper bound for daily horizons

Project Goals & Success Thresholds:

- **Discrete Classification – Accuracy, Precision, Recall**
 - **Baseline:** Random Walk (50%)
 - **Success:** > 55% Accuracy
- **Continuous – RMSE, R^2**
 - **Baseline:** Naive "Always 0" Strategy
 - **Success:** < 0.01 RMSE, R^2 near zero

ECONSTOR
Make Your Publications Visible.

A Service of
ZBW
Leibniz-Informationszentrum
Wirtschaft
Leibniz-Information Science
Service

Zhong, Xiao; Enke, David

Article
Predicting the daily return direction of the stock market using hybrid machine learning algorithms

Financial Innovation

Provided in Cooperation with:
Springer Nature

Suggested Citation: Zhong, Xiao; Enke, David (2019) : Predicting the daily return direction of the stock market using hybrid machine learning algorithms, Financial Innovation, ISSN 2199-4730, Springer, Heidelberg, Vol. 5, Iss. 1, pp. 1-20, <https://doi.org/10.1186/s40854-019-0138-0>

מסמכים מקוים/למודים/תואר ראשון/תואר שני/מסמכים/Workshop in
act/src/data

The documents of EconStor may be saved and copied for your personal and scholarly purposes.
You are not to copy documents for public or commercial purposes, to exhibit the documents publicly, to make them publicly available on the Internet, or to distribute or otherwise use the documents in public.
If the documents have been made available under an open content license (especially Creative Commons License), you may exercise further usage rights as specified in the relevant license.

<https://nbn-resolving.org/urn:nbn:de:hbz:5:1-63864-p0011-9>

www.econstor.eu

Research on the Optimal Prediction Model of Stock Returns of FAANG+M

Yijun Xia
Business School, Soochow University, Suzhou, Jiangsu, 215006, China
2210410019@stu.suda.edu.cn

Abstract. As machine learning continues to mature and advance, stock prediction has been extensively discussed across diverse fields of study. Compared to pure price, stock returns seems to matter more, for people tend to always attach most importance to the chance of winning revenue. However, research that deals with relative return forecasting and focuses on a specific collection of stocks is still limited. In order to promote the wider application in real life and facilitate the process of trading in stock market, this paper examines the optimal prediction model of stock returns by gathering stock price data from 2015 to 2024 in six enterprises (FAANG+M), adding time series parameters based on existing information, selecting NASDAQ-100 Index as the baseline parameter to eliminate market fluctuations, and constructing separate three regression models (Linear Regression, Random Forest and Extreme Gradient Boosting). Overall, Linear Regression possesses the optimal prediction outcomes, while Random Forest takes second place and XGBoost the last. In other words, in the particular context of this paper, there seems to be a positive relationship between model simplicity and result quality. The outcome deviates from conventional expectations and can be partly explained by the complexity of the models themselves and the overfitting issues that ensue. To tackle this problem, regularization and early stopping technology can be further employed to optimize model performance.

Keywords: Stock Return, Linear Regression, Random Forest, XGBoost

1 Introduction

The prediction of stock returns has been garnering significant attention during the last few decades as more multifaceted and comprehensive methods are continuously popping up. In the commonly accepted Effective Market Theory, however, all market information has been embodied in the corresponding stock price, and any disparities will be bridged immediately due to the rapid reactions from market participants. Therefore, the price follows the Random Walk Hypothesis and is highly challenging to predict. However, since the invention and application of machine learning skills, scholars from relative fields have

© The Author(s) 2025
Y. Xia (ed.), Proceedings of the 2025 3rd International Conference on Image, Algorithms, and Artificial Intelligence (ICIAAI 2025), Advances in Computer Science Research 122.
https://doi.org/10.22919/978-94-4465-423-3_71

Feature Engineering



BASIC (OHLC)

Open High Low Close

MOVING AVERAGES

MA20 MA50 MA200

Dist_MA50 Dist_MA200

DIVIDENDS

Dividends

RETURNS

Return Log_Return

VOLATILITY

Vol20

STOCK SPLITS

Stock Splits

PEER DYNAMICS (LAG-1)

AAPL (C, V, LR) MSFT (C, V, LR)

AMZN (C, V, LR)

GOOG (C, V, LR)

Feature Engineering

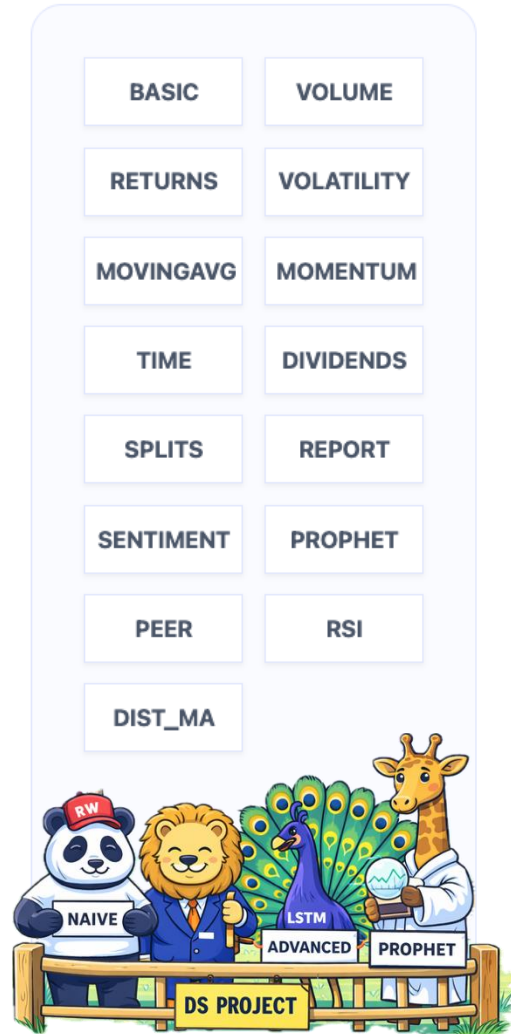


- Interaction Variables
- Cleaning
- Standardization
- Encoding
- Automated time frames

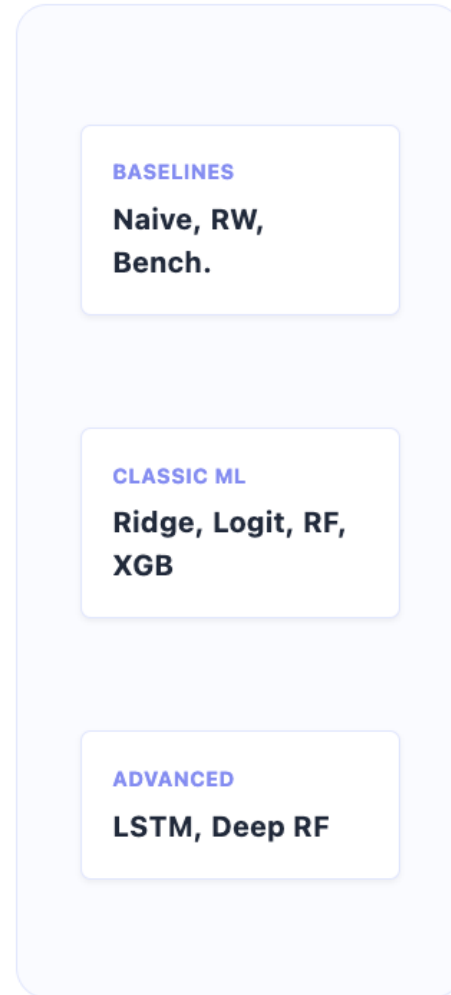
BASIC (OHLC) Open High Low Close	VOLUME DYNAMICS Volume Vol_change Vol_MA20 Rel_vol	RETURNS Return Log_Return	VOLATILITY Vol20
MOVING AVERAGES MA20 MA50 MA200 Dist_MA50 Dist_MA200	MOMENTUM & RSI MACD MACD_Signal MACD_Hist RSI	TIME COMPONENTS Month Day_sin Day_cos	STOCK SPLITS Stock Splits
DIVIDENDS Dividends	CORPORATE REPORTS Days To Nearest Report	META-PROPHET Prophet_Bin Prophet_Cont	PEER DYNAMICS (LAG-1) AAPL (C, V, LR) MSFT (C, V, LR) AMZN (C, V, LR) GOOG (C, V, LR)
SENTIMENT (NLP - FINBERT LAGS) mean_lag1 count_lag1 market_lag1 trend_lag1 std_lag1 Score ma_7d mom_3d vol_7d		MACRO & MARKET INDICATORS Nasdaq_100 Treasury_10Y VIX_Index VIX_MA20 VIX_Gap NVDA_Leader	

“Model Zoo” Experiment- Framework

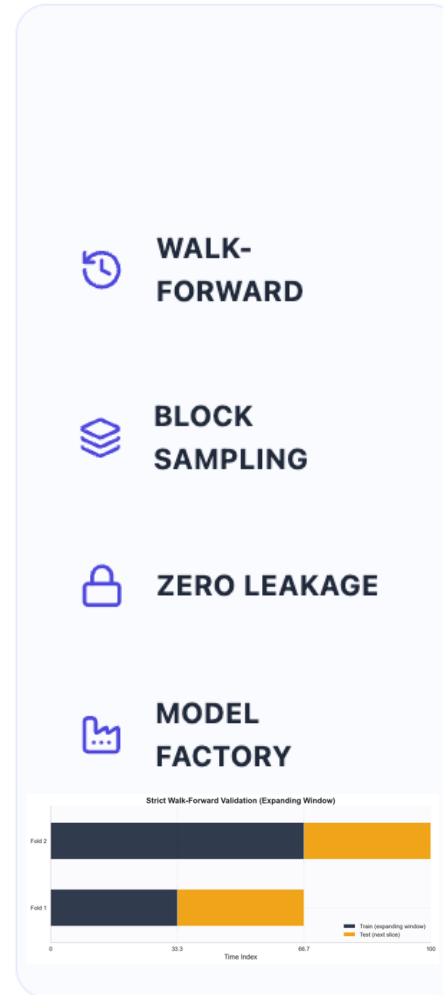
01. INPUT REGISTRY



02. MODEL ZOO



03. PROTOCOL



04. TARGET



“Model Zoo” Experiment- Results



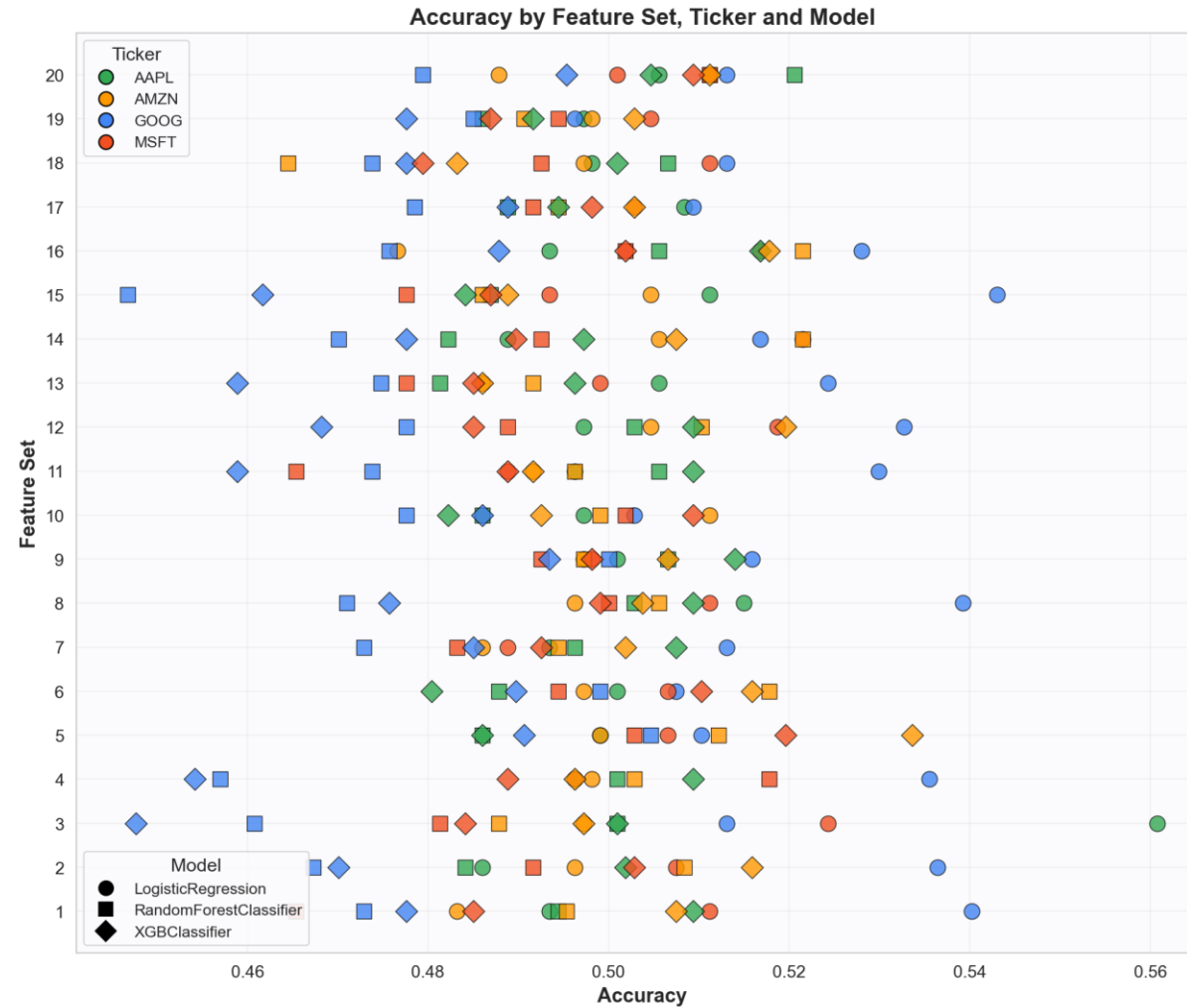
- Continuous run



“Model Zoo” Experiment- Results

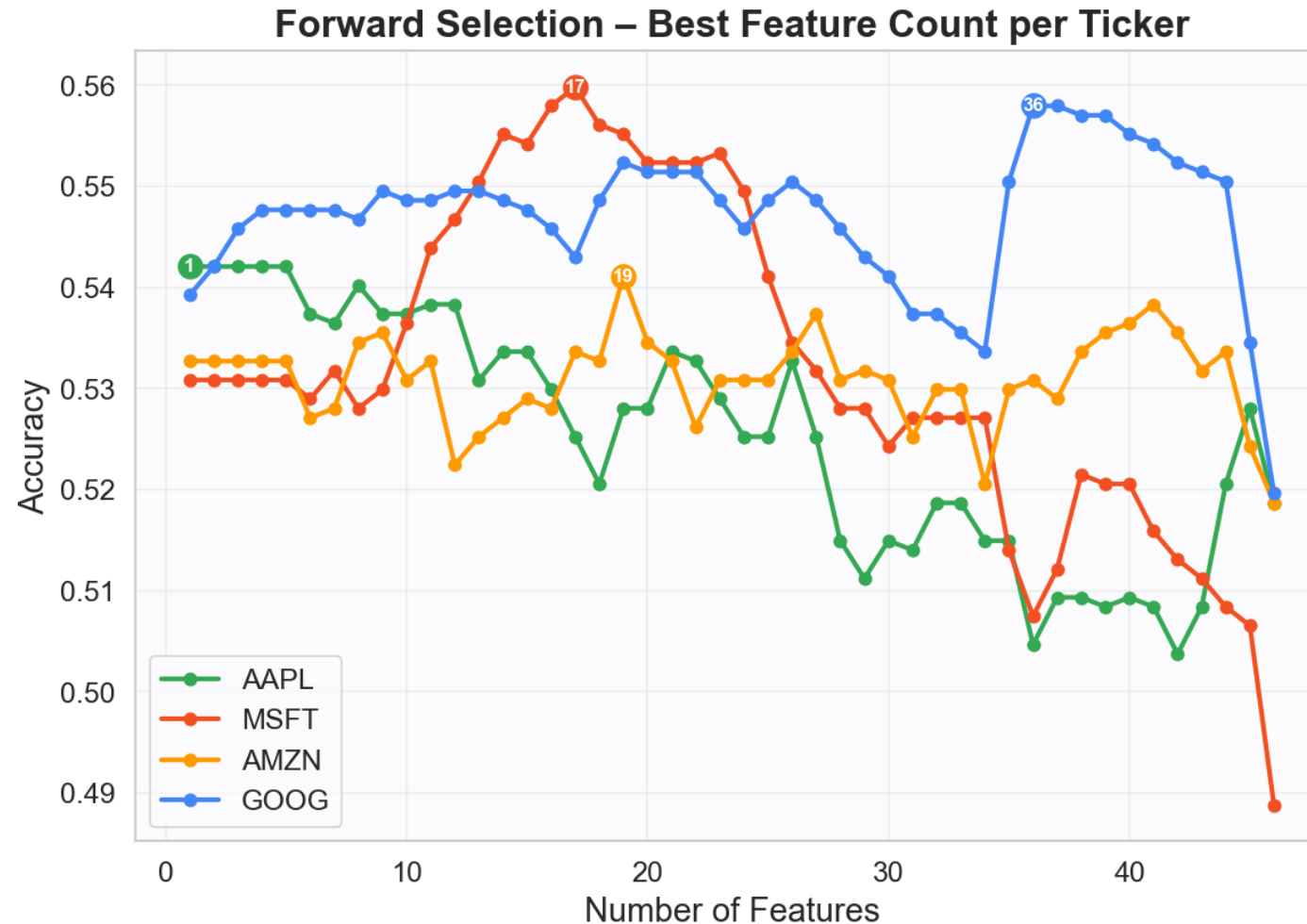


- Discrete run



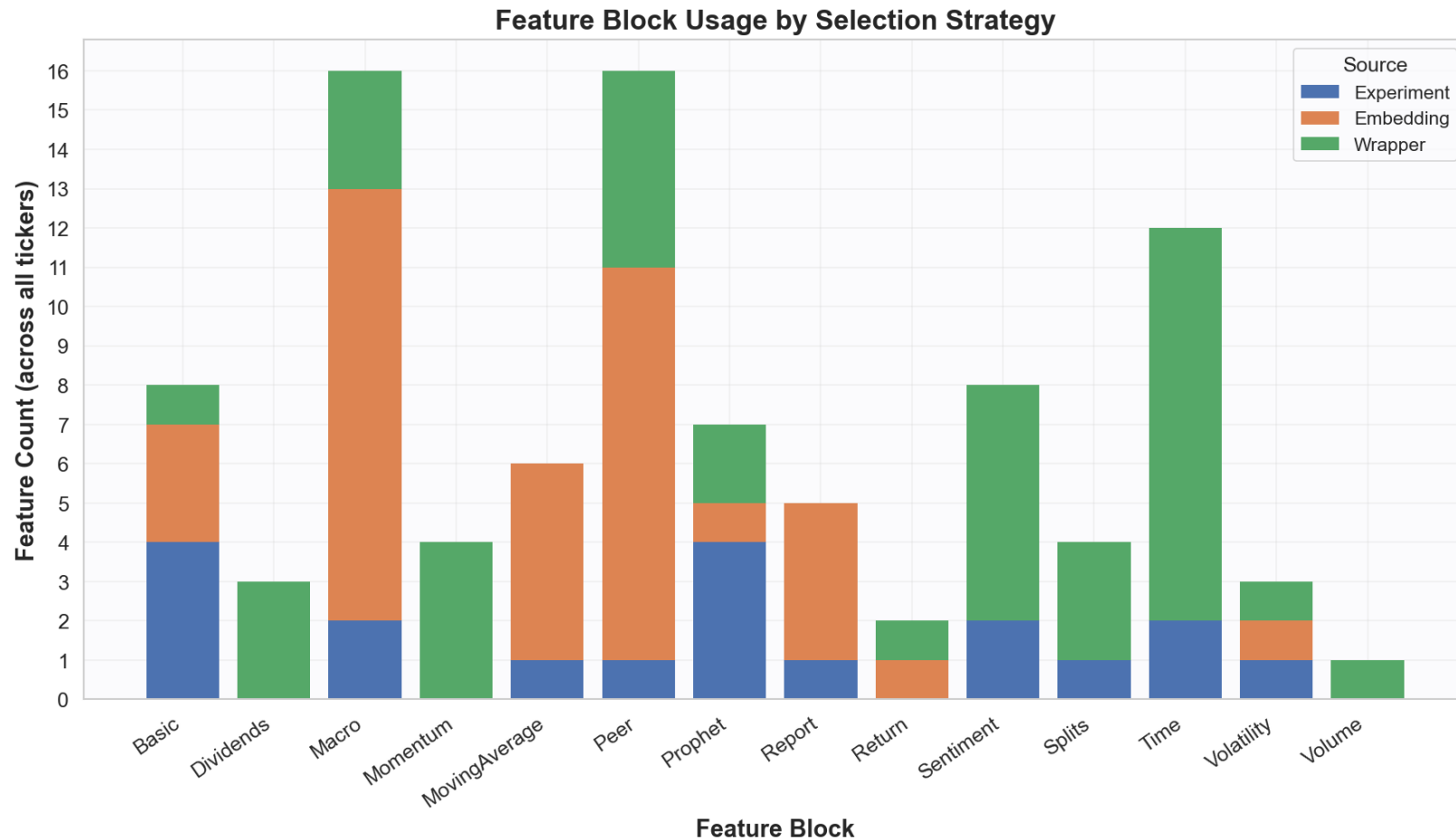
Feature Selection

- Wrapper method – forward selection



Feature Selection

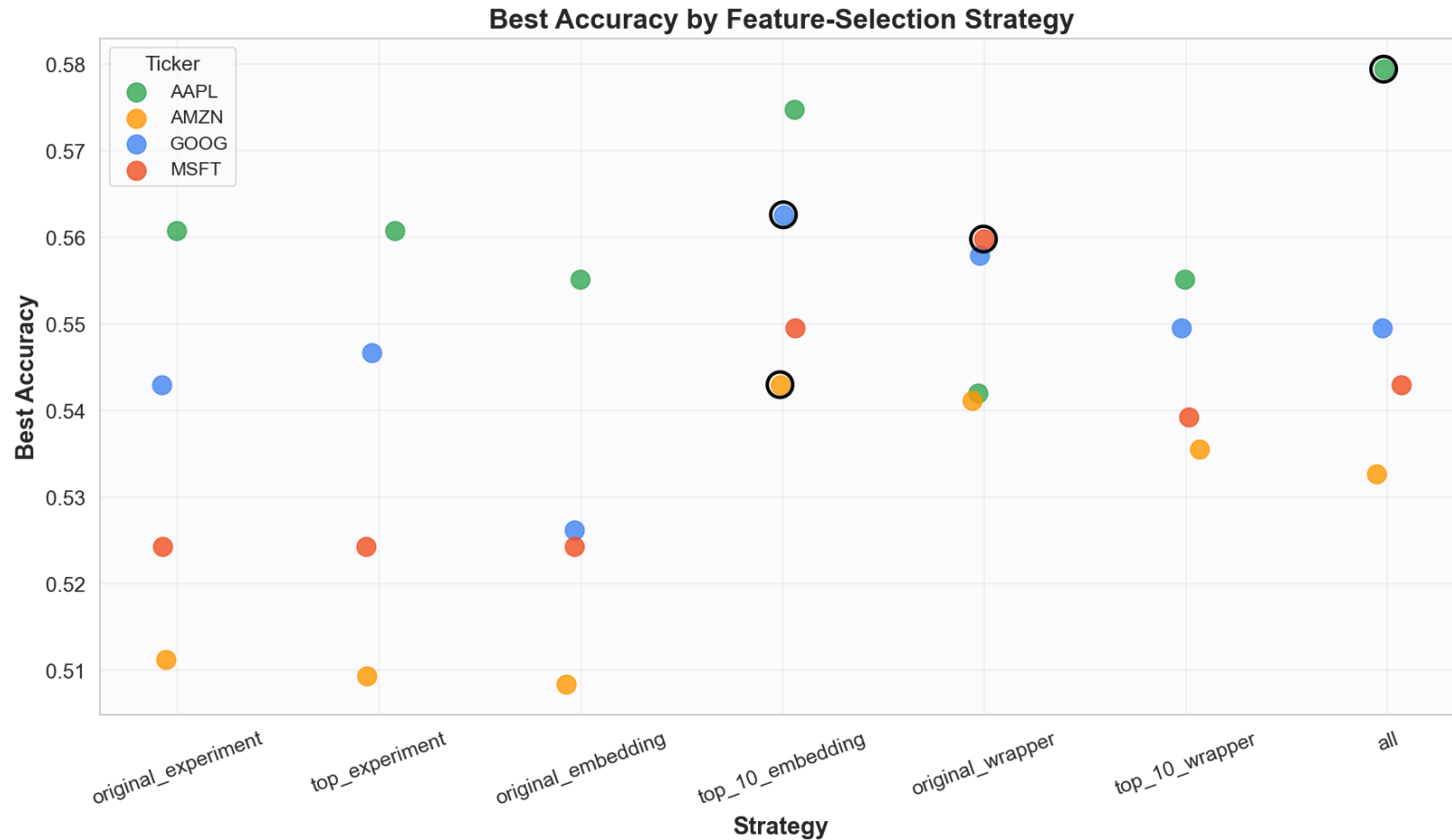
- Wrapper method – forward selection
- Comparing top features from strategies



Feature Selection

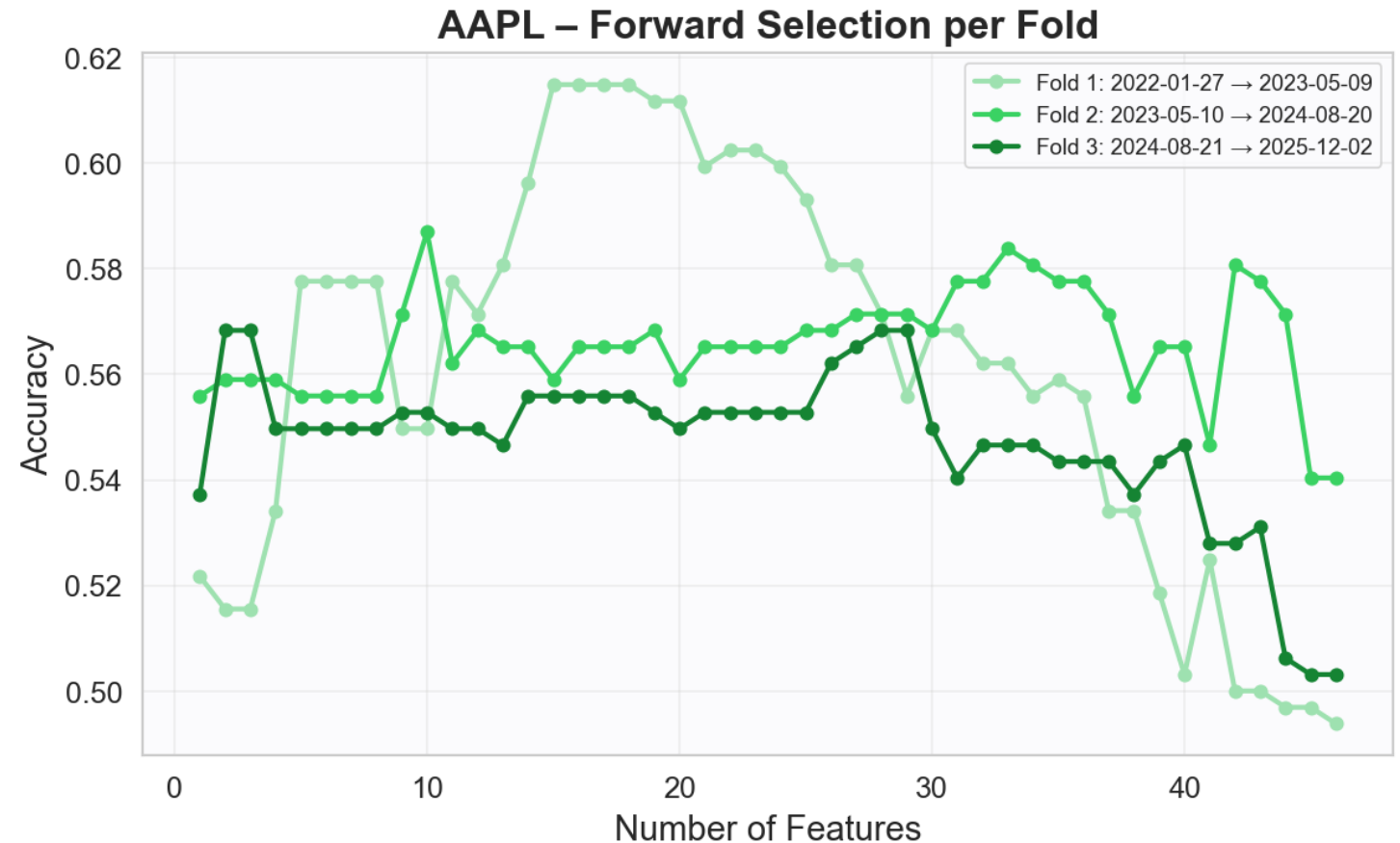
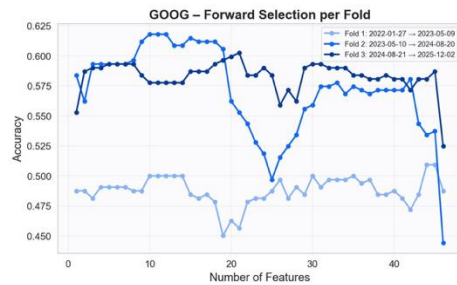
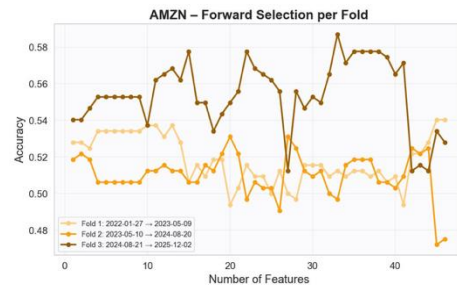
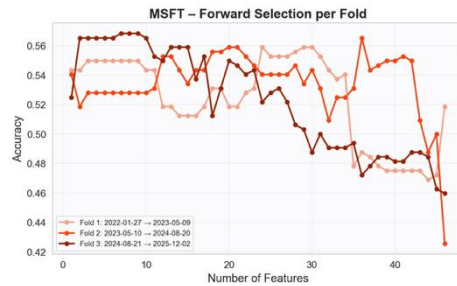


- Wrapper method – forward selection
- Comparing top features from strategies

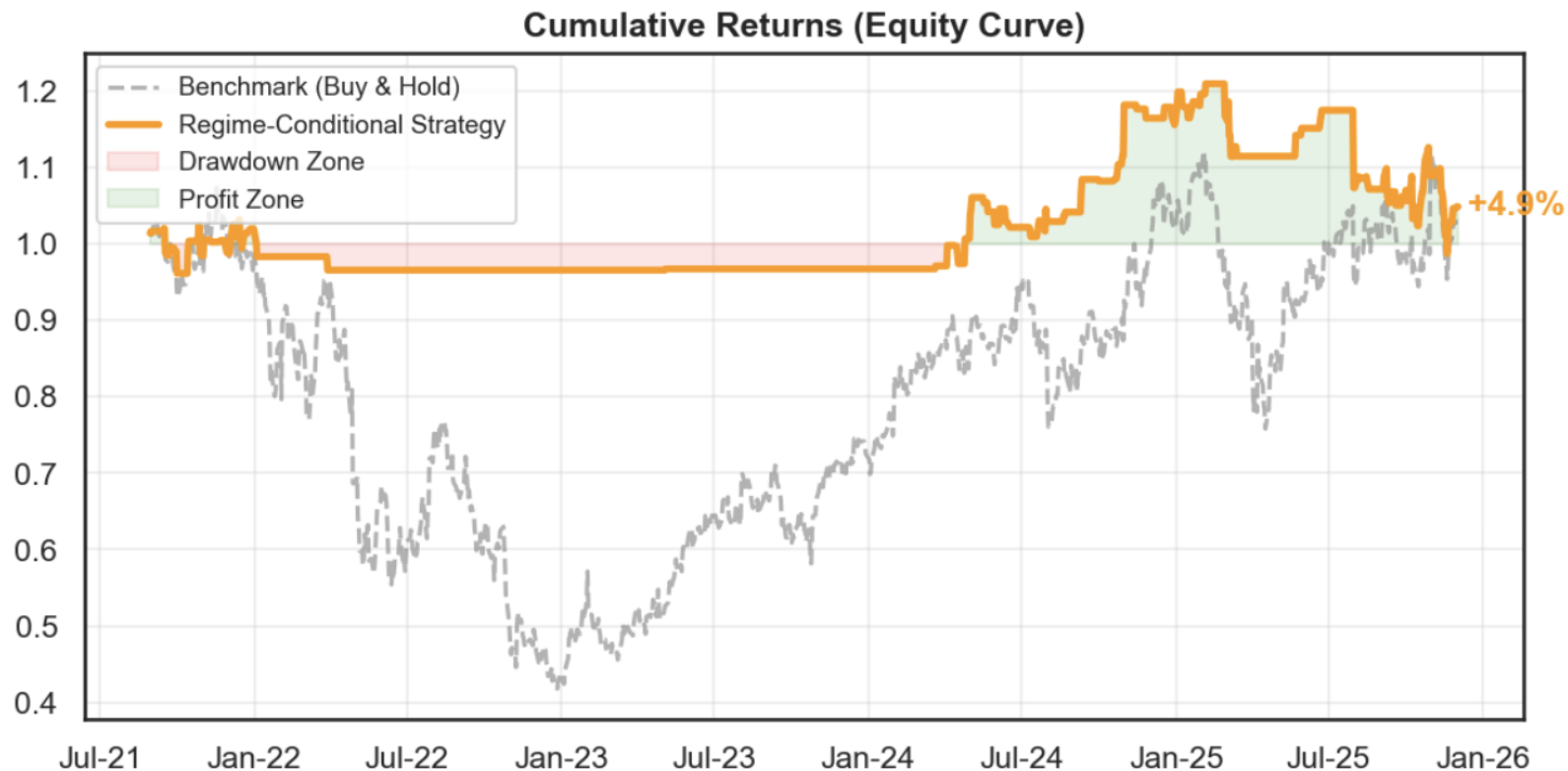


Feature Selection

- Wrapper method – forward selection
- Comparing top features from strategies
- Per fold

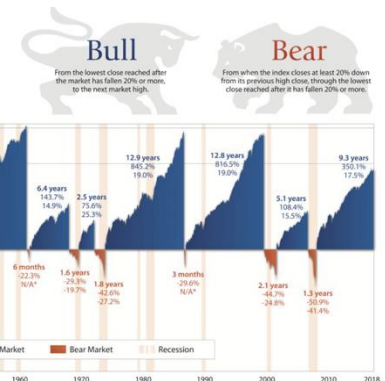


Future Direction: Bull And Bear Regimes



This chart shows historical performance of the S&P 500 index throughout the U.S. Bull and Bear Markets from 1926 through June 2018. Although past performance is no guarantee of future results, we believe looking at the history of the market's expansions and recessions helps to gain a fresh perspective on the benefits of investing for the long term.

- The average Bull Market period lasted 6.1 years with an average cumulative total return of 43%.
- The average Bear Market period lasted 1.4 years with an average cumulative loss of -41%.



Project status- Finished tasks

Data Preparation:

- Data Access
- EDA
- Feature Engineering

Evaluation:

- Benchmarks and Success Criteria

Model Selection:

- Experiment
- Feature Selection

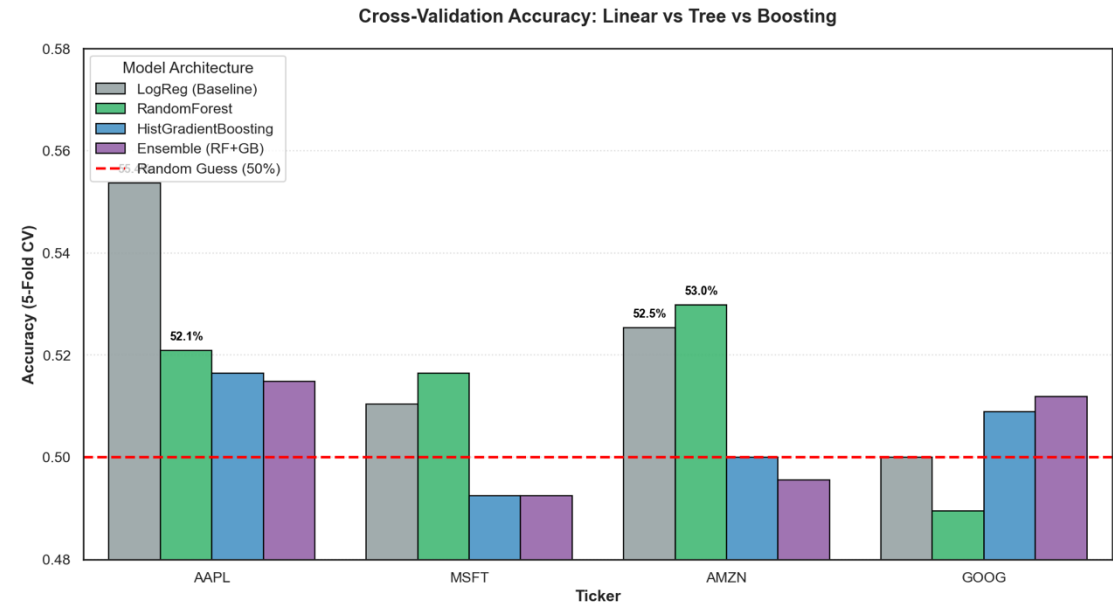
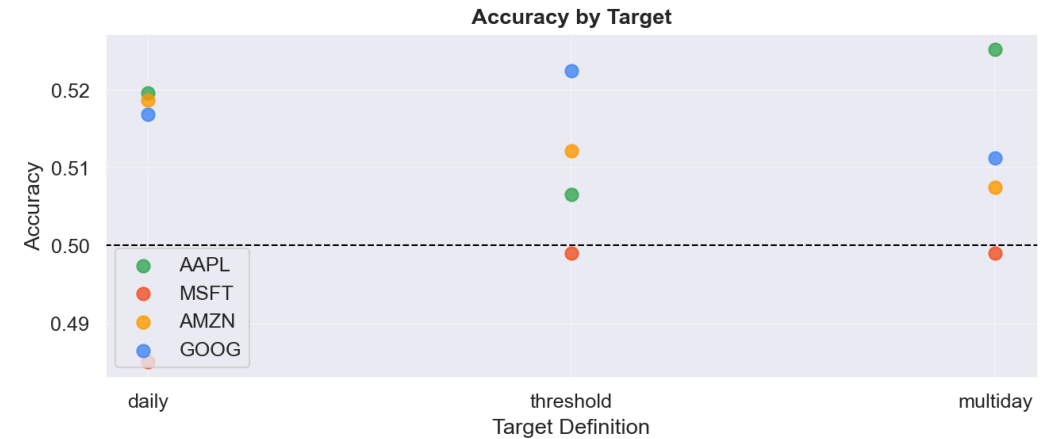
Project status – Future Directions

Remaining Tasks Before Submission:

- Code debugging and cleaning
- Notebook Order and Plotting Improvements
- Explainability
- Discussion

Additional Deep Dive Directions:

- Focus on RandomForest
- Target horizon exploration
- Bull & Bear regimes
- Continuous run against baseline
- Explore difference between folds



Thank you, Questions?

